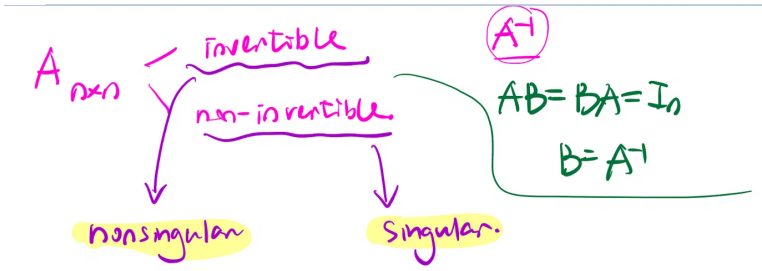
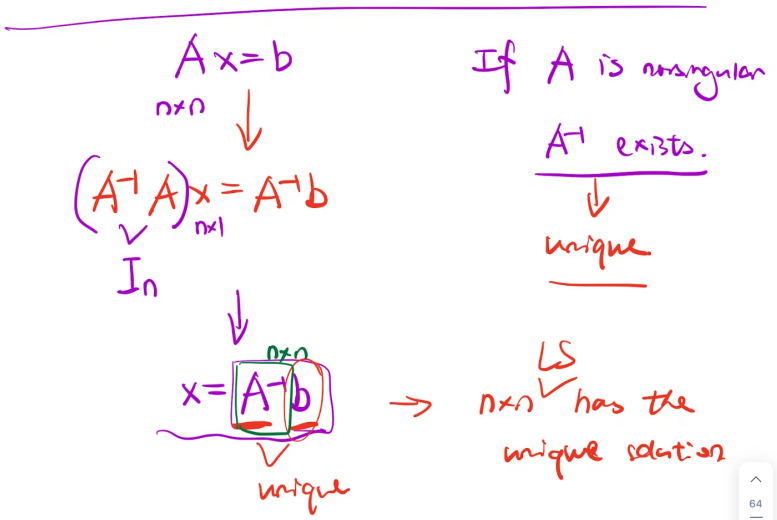
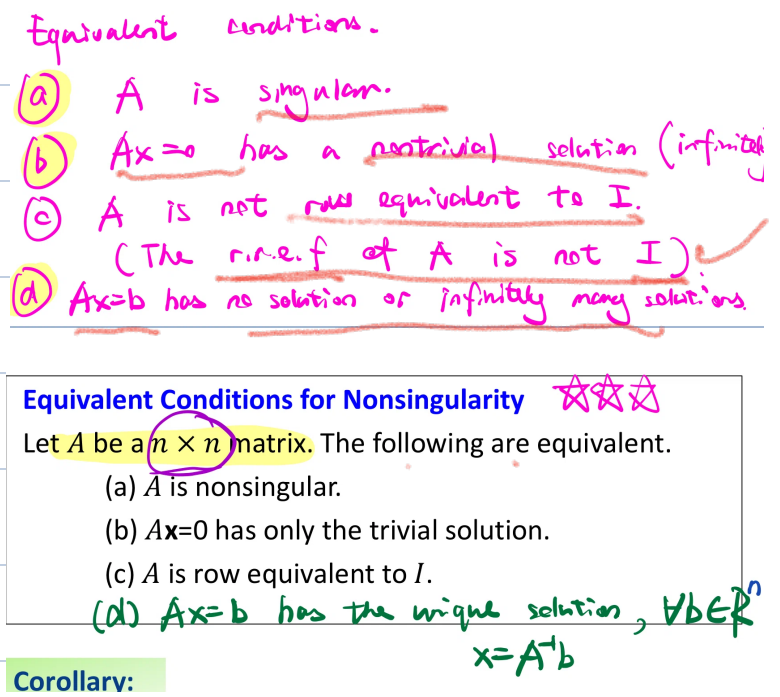
1. Inverse and Singularity of matrix

2. Elementary matrix and Gauss-Jordan reduction for inverse







2. (a) The matrix A is non-singular. Show that A^T is also non-singular.

   (b) The matrix A is non-singular and symmetric. Show that A^{-1} is also symmetric.

4. Let A and B be $n \times n$ matrices and let $C = A - B$. Show that if $A\mathbf{x}_0 = B\mathbf{x}_0$ and $\mathbf{x}_0 \neq 0$, then C must be singular.

Definition

An **elementary matrix** is one that can be obtained from the identity matrix I_n through a single elementary row operation.

Example 1

Summary:

Type I: $E: R_i \leftrightarrow R_j$, $E^{-1}: R_i \leftrightarrow R_j$

Type II: $E: \alpha R_i$, $\alpha \neq 0$ $E^{-1}: \frac{1}{\alpha} R_i$

Type III: $E: R_j + \alpha R_i$ $E^{-1}: R_j - \alpha R_i$

- In general, suppose E is an $n \times n$ elementary matrix. We can think of E as being obtained from I by either a row operation or a column operation.
- If A is an $n \times r$ matrix, **pre-multiplying** A by E (i.e., EA) has the effect of performing that same row operation on A.
- If B is an $m \times n$ matrix, **post-multiplying** B by E (i.e., BE) is equivalent to performing that same column operation on B.

Gauss-Jordan Elimination for finding the Inverse of a Matrix

Let A be an $n \times n$ matrix.

1. Adjoin the identity $n \times n$ matrix I_n to A to form the matrix $[A \mid I_n]$.
2. Compute the reduced echelon form of $[A \mid I_n]$.

   If the reduced echelon form is of the type $[I_n \mid B]$, then B is the inverse of A.

   If the reduced echelon form is not of the type $[I_n \mid B]$, i.e. the left-hand side $n \times n$ matrix is not I_n, then A has no inverse.

3. Given

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 3 \\ 1 & 0 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 1 & 3 \\ 2 & 2 & 6 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 & 4 \\ 0 & -1 & -3 \\ 2 & 2 & 6 \end{bmatrix}$$

(a) Find an elementary matrix E_1 such that $E_1 A = B$.

(b) Find an elementary matrix E_2 such that $E_2 B = C$.

(c) Find A^{-1}.

(d) Find C^{-1} utilising E_1^{-1} and E_2^{-1}.

(e) Utilize A^{-1} obtained in (c) to solve the linear system $A\mathbf{x} = \mathbf{b}$, where $\mathbf{b} = [1, 2, 2]^T$.